

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT:- AI PRACTICAL NO: 1

AIM: Breadth First Search & Iterative Depth First Search in Python.

- A) Implement the Breadth First Search algorithm to solve a given problem.
- B) Implement the Iterative Depth First Search algorithm to solve the same problem.
- C) Compare the performance and efficiency of both algorithms.

OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/mvlui/Documents/TY089CS/AI/prac 1.py =====
Source: Borivali
Destination: Dadar Station
-----
BFS Traversal:
Borivali
Kandivali
Dahisar
Malad
Goregaon
Mira Road
Jogeshwari
Andheri
Bandra
Dadar West
Dadar Station

Shortest Path:
Borivali -> Dahisar -> Mira Road -> Bandra -> Dadar Station
Source      : Borivali
Destination : Dadar Station
-----
DFS Traversal:
Borivali
Kandivali
Malad
Jogeshwari
Bandra
Dadar Station

Path Found:
Borivali -> Kandivali -> Malad -> Jogeshwari -> Bandra -> Dadar Station
===== BFS vs DFS Comparison =====
```

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Kandivali
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Path Found:
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===== BFS vs DFS Comparison =====

BFS Path: Borivali -> Dahisar -> Mira Road -> Bandra -> Dadar Station
DFS Path: Borivali -> Kandivali -> Malad -> Jogeshwari -> Bandra -> Dadar Station

Execution Time:
BFS Time: 1.1444091796875e-05
DFS Time: 5.7220458984375e-06

Conclusion:
BFS gives shortest path, DFS may not but uses less memory.
>>>
```

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ROLL NO: 089